

The Turning Effect — Paper A

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Always convert centimetres to metres **before** you multiply, and give every moment the unit **N m**.
- The mark for each question is shown in square brackets on the right.
- Reminder: **moment = force × perpendicular distance from the pivot**.

SECTION 1 – THE IDEA

1. Answer these three parts about the moment of a force.

(a) State what is meant by the **moment of a force**. [1]

(b) Write down the formula for a moment. [1]

(c) State the unit of a moment. [1]

2. Calculate the moment in each case.

(a) A force of 20 N acting 0.5 m from the pivot. [1]

(b) A force of 40 N acting 25 cm from the pivot. [2]

(c) A force of 12 N acting 1.5 m from the pivot. [1]

- 3.** A mechanic uses a spanner with a long handle rather than a short one. Explain why this makes the nut easier to undo. [2]

- 4.** A beam is free to turn about a pivot at its centre.

- (a)** A force pushes down on the right-hand end. State the direction of the turning effect. [1]

- (b)** A force pushes down on the left-hand end. State the direction of the turning effect. [1]

SECTION 2 – CALCULATION

- 5.** Convert each distance into metres, ready for the moment formula.

- (a)** 60 cm [1]

- (b)** 250 cm [1]

- 6.** A force of 25 N acts at a perpendicular distance of 40 cm from a pivot, turning the object clockwise. Calculate the moment and state its direction. [3]

- 7.** The handle of a door is fitted as far as possible from the hinges. Explain why. [2]

8. A child of weight 300 N sits 2 m from the pivot of a see-saw. Calculate the moment of his weight about the pivot. [2]

SECTION 3 – BALANCE

9. State the **principle of moments**. [2]

10. State where the pivot is in each case.

- (a) A door being pushed open. [1]

- (b) A spanner turning a nut. [1]

11. A force of 15 N acts 0.8 m to the left of a pivot. A force of 24 N acts 0.5 m to the right of the same pivot.

- (a) Calculate the anticlockwise moment. [1]

- (b) Calculate the clockwise moment. [1]

- (c) State, with a reason, whether the beam balances. [2]

12. Explain why the distance in the moment formula must be the **perpendicular** distance from the pivot.

[2]

End of paper. Check every answer has a unit or a form the question actually asked for.